Calculus Practice — Newton's Method for Optimization

8 questions covering the Newton's method update rule, its connection to the second-order Taylor expansion, extension to multiple variables via the Hessian, failure cases, and a comparison with gradient descent. Try all of them before checking the answers below.

Practice Questions

Q1 Apply one step of Newton's method to minimize $f(x) = x^2 - 6x + 5$, starting from $x_0 = 0$. The update rule is:

$$x_{n+1} = x_n - \frac{f'(x_n)}{f''(x_n)}$$

What is x_1? What do you notice about the result?

Q2 Use Newton's method to minimize $f(x) = x^3 - 3x + 1$, starting at $x_0 = 2$. Compute x_1 and x_2, and compare the rate at which you are converging to the true minimum.

Q3 Show that Newton's method always reaches the **exact minimum of any quadratic** $f(x) = ax^2 + bx + c$ (with $a > 0$) in **exactly one step**, regardless of the starting point x_0.

Q4 Compute the **Hessian matrix** H of:

$$f(x, y) = 2x^2 + 4y^2 - 3x + y$$

(The Hessian is the matrix of all second-order partial derivatives: $H_{ij} = \dfrac{\partial^2 f}{\partial x_i \partial x_j}$*.)*

Q5 Using the Hessian from Q4, perform one step of Newton's method in 2D, starting from $(x, y) = (0, 0)$. The update rule is:

$$\mathbf{p}_{n+1} = \mathbf{p}_n - H^{-1} \nabla f(\mathbf{p}_n)$$

What do you notice about where you land?

Q6 Consider $f(x) = x^3 - 3x$, which has a local minimum at $x = 1$ and a local maximum at $x = -1$. Apply one Newton step starting from $x_0 = -0.5$, and explain what went wrong and why.

Q7 The Newton update $x_{n+1} = x_n - \dfrac{f'(x_n)}{f''(x_n)}$ can be derived by minimizing the **second-order Taylor approximation** of f around x_n:

$$Q(x) = f(x_n) + f'(x_n)(x - x_n) + \frac{1}{2} f''(x_n)(x - x_n)^2$$

Show that minimizing $Q(x)$ with respect to x gives exactly the Newton update.

Q8 Gradient descent updates as $x \leftarrow x - \alpha f'(x)$, while Newton's method updates as $x \leftarrow x - \dfrac{f'(x)}{f''(x)}$. Interpret the role of $\dfrac{1}{f''(x)}$ in Newton's method as an **automatic learning**

rate, and describe two scenarios where Newton's method is preferred over gradient descent and two where gradient descent is preferred.

Challenge Problems

Q9 Newton's method is applied to minimize $f(x) = x^p$ for an even integer $p \geq 2$, starting from $x_0 \neq 0$.

(a) Show that the Newton update simplifies to:

$$x_{k+1} = \frac{p-2}{p-1}\, x_k$$

(b) Hence write down the closed form x_k in terms of x_0, p, and k.

(c) For which value of p does Newton's method converge to the minimum in **exactly one step** from any x_0? Explain why this is consistent with Q3.

(d) For $p = 4$, the method converges, but is it **linear** or **quadratic**? The standard theorem guarantees quadratic convergence near a minimum x^* where $f''(x^*) \neq 0$. Compute $f''(0)$ for general $p > 2$ and use this to explain why quadratic convergence breaks down, and what replaces it.

Q10 For $f(x) = x^3 - 3x$, define the **Newton map** $N(x) = x - \dfrac{f'(x)}{f''(x)}$, so that the Newton iteration is simply $x_{k+1} = N(x_k)$.

(a) Show that $N(x) = \dfrac{x^2+1}{2x}$.

(b) The fixed points of N are the points satisfying $N(x^*) = x^*$. Find all fixed points of N and show they are exactly the critical points of f. What does this tell you about what Newton's method converges to?

(c) Compute $N'(x)$ and evaluate $|N'(x^*)|$ at each fixed point. Using the fact that near a fixed point the error satisfies:

$$e_{k+1} \approx N'(x^*)\, e_k + \frac{N''(x^*)}{2}\, e_k^2$$

prove that Newton's method exhibits **quadratic convergence** near each fixed point, and find the constant in the leading error term.

(d) At any point where $f''(x) = 0$, the Newton step is undefined. Find where this occurs for f, classify this point geometrically, and show explicitly that the second-order Taylor approximation of f at this point **degenerates into a straight line** — explaining precisely why no Newton step can be taken.

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Answers & Solutions

A1. $f'(x) = 2x - 6$, $f''(x) = 2$.

$$x_1 = 0 - \frac{2(0) - 6}{2} = 0 - \frac{-6}{2} = 3$$

Newton's method lands **exactly on the minimum** ($x = 3$, where $f'(3) = 0$) in a single step, regardless of where we started. This is no coincidence — Q3 shows this always happens for quadratics.

A2. $f'(x) = 3x^2 - 3$, $f''(x) = 6x$.

Step 1:

$$x_1 = 2 - \frac{3(4) - 3}{6(2)} = 2 - \frac{9}{12} = 2 - 0.75 = 1.25$$

Step 2:

$$x_2 = 1.25 - \frac{3(1.5625) - 3}{6(1.25)} = 1.25 - \frac{1.6875}{7.5} = 1.25 - 0.225 = 1.025$$

The true minimum is at $x = 1$ (where $f'(x) = 0$). The errors after each step are $|x_1 - 1| = 0.25$, $|x_2 - 1| = 0.025$ — the error shrank by a factor of roughly $10\times$ in one step. With gradient descent (for comparison), the error typically shrinks by a constant factor each step; with Newton's method it shrinks **quadratically**: each step roughly *squares* the previous error. That's why it only takes a handful of steps to get very close.

A3. $f'(x) = 2ax + b$, $f''(x) = 2a$. Applying the Newton update from any x_0:

$$x_1 = x_0 - \frac{2ax_0 + b}{2a} = x_0 - x_0 - \frac{b}{2a} = -\frac{b}{2a}$$

This is the exact minimizer of f (where $f'(x) = 0$), independent of x_0. Geometrically: a quadratic has the same curvature everywhere, so the local quadratic approximation (which Newton minimizes) *is* the function itself — there is no approximation error.

A4. Computing all second-order partials:

$$\frac{\partial^2 f}{\partial x^2} = 4, \quad \frac{\partial^2 f}{\partial y^2} = 8, \quad \frac{\partial^2 f}{\partial x \partial y} = 0$$

$$H = \begin{pmatrix} 4 & 0 \\ 0 & 8 \end{pmatrix}$$

Since f is quadratic in x and y with no cross terms, the Hessian is diagonal — the two directions are completely decoupled.

A5. $\nabla f(x, y) = (4x - 3,\ 8y + 1)$. At $(0, 0)$: $\nabla f = (-3,\ 1)$.

H is diagonal so its inverse is immediate: $H^{-1} = \begin{pmatrix} 1/4 & 0 \\ 0 & 1/8 \end{pmatrix}$.

$$\mathbf{p}_1 = \begin{pmatrix}0\\0\end{pmatrix} - \begin{pmatrix}1/4 & 0\\ 0 & 1/8\end{pmatrix}\begin{pmatrix}-3\\1\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix} - \begin{pmatrix}-3/4\\1/8\end{pmatrix} = \begin{pmatrix}3/4\\-1/8\end{pmatrix}$$

We land exactly on the global minimum in one step (verifiable by setting $\nabla f = 0$: $x = 3/4$, $y = -1/8$). Again, because f is quadratic, Newton's method is exact — the Hessian captures the full curvature of the function with no approximation.

A6. $f'(x) = 3x^2 - 3$, $f''(x) = 6x$. At $x_0 = -0.5$:

$$f'(-0.5) = 3(0.25) - 3 = -2.25, \quad f''(-0.5) = 6(-0.5) = -3$$

$$x_1 = -0.5 - \frac{-2.25}{-3} = -0.5 - 0.75 = -1.25$$

This is moving toward $x = -1$, the **local maximum** of f, not the minimum at $x = 1$.

Why it went wrong: Newton's method finds where $f'(x) = 0$, but it cannot distinguish between a minimum and a maximum on its own — it is simply finding a root of f'. The sign of f'' tells us which it is: at $x = -1$, $f''(-1) = -6 < 0$, which means f' is *decreasing* there — a local maximum of f. Newton's method was drawn toward it because x_0 was in the basin of $x = -1$ rather than $x = 1$. This is the same basin-of-attraction issue from gradient descent, but with an added danger: a negative Hessian actively steers Newton's method *toward* saddle points and maxima.

A7. $Q(x)$ is a quadratic in $(x - x_n)$. Taking the derivative with respect to x and setting it to zero:

$$Q'(x) = f'(x_n) + f''(x_n)(x - x_n) = 0$$
$$x - x_n = -\frac{f'(x_n)}{f''(x_n)}$$
$$x = x_n - \frac{f'(x_n)}{f''(x_n)}$$

This is exactly the Newton update. In other words: Newton's method fits a parabola to f at the current point (matching its value, slope, and curvature), finds the exact bottom of that parabola, and jumps there. The better f is approximated by a quadratic locally, the more accurate the step.

A8. The factor $\dfrac{1}{f''(x)}$ plays the role of an **automatic, curvature-adapted learning rate**:

- Where $f''(x)$ is large (the function curves steeply), $\dfrac{1}{f''(x)}$ is small, so Newton takes a short, careful step.
- Where $f''(x)$ is small (the function is nearly flat), $\dfrac{1}{f''(x)}$ is large, so Newton takes a bold step. Gradient descent with a fixed α cannot do this — it takes equally sized steps regardless of curvature, which is why it crawls in flat regions and can overshoot in steep ones.

Newton's method is preferred when: - The problem is small-scale and computing the Hessian is affordable. - Very fast convergence is needed (e.g., in classical optimisation where function evaluations are expensive).

Gradient descent is preferred when: - The number of parameters is large (e.g., a neural network with millions of weights) — inverting an $n \times n$ Hessian costs $O(n^3)$ operations, making it completely impractical at scale. - Stochastic or mini-batch updates are used (the Hessian is not well-defined on a mini-batch), which is the standard regime in deep learning.

Challenge Problem Answers

A9.

(a) $f'(x) = px^{p-1}$ and $f''(x) = p(p-1)x^{p-2}$. The Newton step is:

$$x_{k+1} = x_k - \frac{px_k^{p-1}}{p(p-1)x_k^{p-2}} = x_k - \frac{x_k}{p-1} = x_k\left(1 - \frac{1}{p-1}\right) = \frac{p-2}{p-1}\,x_k$$

(b) Applying this repeatedly with constant ratio $r = \dfrac{p-2}{p-1}$:

$$x_k = x_0\left(\frac{p-2}{p-1}\right)^k$$

(c) For $p = 2$: $r = 0$, so $x_1 = 0$ regardless of x_0 — the minimum is reached in one step. This is consistent with Q3: $f(x) = x^2$ is a quadratic, and Newton's method is exact on quadratics in one step.

(d) For $p = 4$: $r = 2/3$, so $x_k = x_0(2/3)^k$. This is a geometric sequence, which is the hallmark of **linear convergence** — each step reduces the error by a fixed multiplicative factor, not by squaring it.

The standard theorem guarantees quadratic convergence at a minimum x^* only when $f''(x^*) \neq 0$. But for $f(x) = x^p$ with $p > 2$:

$$f''(0) = p(p-1) \cdot 0^{p-2} = 0$$

The second derivative **vanishes** at the minimum. This is a degenerate minimum — the function is "flatter than a parabola" there. The Newton map reduces to $N(x) = \frac{p-2}{p-1}x$, which is a linear contraction with $N'(0) = \frac{p-2}{p-1}$ — a nonzero constant — confirming linear convergence. As $p \to \infty$ the ratio $\to 1$ and convergence slows further, since the minimum becomes increasingly flat.

A10.

(a) $f'(x) = 3x^2 - 3$, $f''(x) = 6x$.

$$N(x) = x - \frac{3x^2-3}{6x} = \frac{6x^2 - 3x^2 + 3}{6x} = \frac{3x^2+3}{6x} = \frac{x^2+1}{2x}$$

(b) Setting $N(x^*) = x^*$:

$$\frac{(x^*)^2+1}{2x^*} = x^* \implies (x^*)^2 + 1 = 2(x^*)^2 \implies (x^*)^2 = 1 \implies x^* = \pm 1$$

The critical points of f satisfy $f'(x) = 3(x^2-1) = 0$, giving $x = \pm 1$ — identical. This is not a coincidence: fixed points of the Newton map are always exactly the zeros of f', which are the critical points of f. Crucially, Newton's method makes no distinction between minima and maxima — it converges to **any** critical point it is drawn towards, including $x = -1$, which is a local maximum of f.

(c) $N'(x) = \dfrac{d}{dx}\left[\dfrac{x^2+1}{2x}\right] = \dfrac{2x \cdot 2x - (x^2+1)\cdot 2}{4x^2} = \dfrac{x^2-1}{2x^2}$

At both fixed points $x^* = \pm 1$: $N'(\pm 1) = \dfrac{1-1}{2} = 0$.

Since $N'(x^*) = 0$, the linear error term vanishes and the error expansion becomes:

$$e_{k+1} \approx \frac{N''(x^*)}{2}\, e_k^2$$

This is **quadratic convergence** — the error is squared at every step. Computing $N''(x) = \dfrac{d}{dx}\left[\dfrac{x^2-1}{2x^2}\right] = \dfrac{1}{x^3}$, so $N''(1) = 1$ and the leading error term is:

$$e_{k+1} \approx \frac{1}{2}\, e_k^2$$

So if the error is 0.01 at one step, it is roughly 0.00005 at the next — confirming the dramatic acceleration seen in Q2. The reason $N'(x^*) = 0$ always holds at non-degenerate critical points of f is itself a theorem: it is equivalent to the fact that $f''(x^*) \neq 0$ (a non-degenerate extremum), as verified by Q9 where this fails and convergence degrades to linear.

(d) $f''(x) = 6x = 0$ at $x = 0$. Since f'' changes sign there ($f'' < 0$ for $x < 0$ and $f'' > 0$ for $x > 0$), this is an **inflection point** of f — the curve changes from concave to convex.

The second-order Taylor approximation of f around $x = 0$ is:

$$Q(x) = f(0) + f'(0)(x-0) + \tfrac{1}{2}f''(0)(x-0)^2 = 0 + (-3)x + 0 = -3x$$

The x^2 term has coefficient $\frac{1}{2}f''(0) = 0$, so $Q(x)$ is a **straight line**, not a parabola. A line has no minimum (it descends to $-\infty$), so Newton's method has no finite step to take — the step size $-f'(0)/f''(0)$ involves division by zero. Geometrically: Newton fits a parabola to the function at each step, but at an inflection point the parabola degenerates to a line, and the method breaks down entirely.